Conclusions: Our results enhance the validity of pCR as a surrogate endpoint for better outcomes in LARC, supporting it as an essential therapeutic aim.

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246P Machine learning-based prediction of recurrence after curative treatment for rectal cancer: A multi-model analysis from a tertiary cancer centre

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Background: Predicting recurrence after curative rectal cancer treatment remains challenging. TNM staging, while widely used, is based mainly on imaging, fails to incorporate treatment response, limiting its ability to stratify individual risk. This is especially true for patients with initially metastatic disease treated with curative intent. We evaluated machine learning (ML) models trained on routinely available clinical, imaging, and pathological variables in a retrospective data from a tertiary cancer centre. Our aim was to assess if ML-based approaches could improve prediction over baseline TNM variables and to compare accuracy across multiple ML models.

Methods: The dataset was divided using an 80:20 stratified split to preserve class balance and SMOTE-based oversampling (k=3) was done to correct the target imbalance of the training dataset. Non-informative features were removed, and feature selection was performed using SelectKBest with ANOVA F-value (f_classif). Four ML models—Logistic Regression(LR), Support Vector Machine (SVM), Random Forest(RF), and XGBoost were optimized using grid search and 5-fold cross-validation across 35 variables. Comparative evaluation was also performed using only three TNM-based variables.

Results: A total of 902 patients who were treated with curative intent between 2017 and 2019 were included in the study. The median follow up was 47.1 months (range: 8.7 − 109.8) and the recurrence rate was 22.3%. With ML-derived features (n=35), XGBoost outperformed all other models, achieving the highest accuracy (0.796), (LR − 0.65, SVM -0.76, RF- 0.75) and AUC (0.705) (LR − 0.61, SVM -0.70, RF- 0.70) with strong specificity (0.943). In contrast, models trained on only three TNM parameters showed markedly lower performance (XGBoost accuracy = 0.464, AUC = 0.577).

Conclusions: In this study machine learning models, particularly XGBoost, substantially enhance recurrence prediction when trained on expanded feature sets beyond TNM parameters. This suggests that integration of routinely available, clinicopathological data through ML may provide clinically meaningful risk stratification, prediction of recurrence risk and may identify those needing close follow up.

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247P Prognostic impact of microsatellite instability and survival disparities in rectal cancer: A SEER-based retrospective analysis

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Background: Microsatellite instability (MSI), a molecular marker of defective DNA mismatch repair (MMR), is observed in ∼7% of rectal cancers. MSI-high (MSI-H) tumors, arising from sporadic or germline MMR deficiency, are highly responsive to immune checkpoint inhibitors. We aimed to evaluate the prognostic significance of MSI in rectal cancer in the era of immunotherapy and to explore demographic disparities in survival using real-world data from the U.S.

Methods: We analyzed data from the Surveillance, Epidemiology, and End Results (SEER) cancer database for patients diagnosed with rectal cancer between 2018 and 2021. We assessed cancer-specific survival (CSS) across MSI subtypes—MSI-H, MSI-low (MSI-L), and microsatellite stable (MSS)—and evaluated survival differences by age, gender, race, and stage. Analyses were performed using R. Kaplan-Meier curves visualized survival outcomes, and group comparisons were done using the log-rank test.

Results: Among 17,487 patients, 3.6% were MSI-H (n=637), 1.9% MSI-L (n=332), and 94.5% MSS (n=16,518). Overall 1-year and 3-year CSS were 90% and 75%, respectively. In metastatic patients, median CSS (mCSS) was 25 months, increasing to 36 months in those with MSI-H tumors. By MSI status, 1- and 3-year CSS were 90.6% and 77.7% for MSI-H, 89.9% and 74.9% for MSS, and 84.8% and 68.6% for MSI-L (p=0.0087). Racial disparities were evident: 1- and 3-year CSS were 92% and 80% in White patients, 88% and 70% in Black patients, and 90% and 75% in Hispanic patients (p<0.0001). Age impacted survival significantly: 1- and 3-year CSS were 91% and 74% in patients <65 years vs. 86% and 61% in those ≥65 (p<0.0001). Females had better long-term survival than males, with 3-year CSS under 80% for both, but significantly higher in females (p<0.0001).

Conclusions: MSI-H status is associated with improved survival, reinforcing its role as a favorable prognostic biomarker in rectal cancer and highlighting the importance of routine MSI testing to guide treatment decisions. Worse outcomes among older adults, males, and Black patients reflect persistent disparities in rectal cancer care. These findings underscore the urgent need to identify and address the drivers of these differences to ensure equitable outcomes.

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248P Young-onset rectal cancer (YORC): A descriptive tertiary center analysis

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Background: The incidence of colorectal cancer in individuals under 50 is rising, a trend referred to as young-onset colorectal cancer (YO-CRC). By 2030, it is estimated that 10% of colon cancers and 25% of rectal cancers will occur in this group, according to the National Cancer Institute. This study aims to investigate clinical and tumor characteristics of young-onset rectal cancer (YORC) in a single tertiary referral center.

Methods: We conducted a retrospective analysis of rectal cancer cases diagnosed at Leuven University Hospitals from 2017 to 2024. Patients were categorized into young-onset (≤50 years) and late-onset (>50 years) groups. Clinical presentation, tumor characteristics, and staging were compared. Mismatch repair (MMR) status was assessed via immunohistochemistry, with microsatellite instability (MSI) confirmed by PCR in MMR-deficient cases. Tumor staging followed RECIST criteria based on MRI and CT imaging. Two-year overall survival (OS) and disease-free survival (DFS) were estimated in non-metastatic cases using Kaplan-Meier analysis.

Results: Of 705 patients, 86 (12.2%) were YORC and 619 (87.8%) were in the control group. Gender and BMI were similar (p > 0.005). YORC patients had better WHO performance status (WHO 0 in 98.8% vs. 82.5%, p = 0.002) and higher rates of MSI tumors (7.0% vs. 1.3%, p = 0.001). They were more symptomatic at diagnosis (91.9% vs. 70.9%, p < 0.001), with more proximal tumors (46.5% vs. 32.5%, p = 0.010), higher metastatic rates (22.1% vs. 11.6%, p = 0.007), lymph node involvement (73.8% vs. 60.7%, p = 0.020), and extramural vascular invasion (40.1% vs. 26.0%, p = 0.0043). T stage and mesorectal fascia involvement did not differ significantly. Among non-metastatic patients, 2-year OS (100% vs. 93.2%, p = 0.250) and DFS (83.8% vs. 84.0%, p = 0.904) were comparable.

Conclusions: YORC patients present with more advanced disease at diagnosis, including higher rates of metastatic (M1), node-positive (N+) disease and extramural vascular invasion (EMVI). YORC patients were more symptomatic and had more MSI tumors. These findings suggest potential biological and clinical differences in YORC. Further research and collaborative efforts are warranted to explore underlying mechanisms and optimize management approaches in YORC.

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